SIMATS ENGINEERING

CSA17 – ARTIFICIAL INTELLIGENCE LAB

Day 1 – Question 8: 8. Depth-First Search (DFS)

Aim: To traverse a graph using Depth-First Search.

Python Program

```
def dfs(graph, node, visited):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)
        for neighbour in graph[node]:
            dfs(graph, neighbour, visited)

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [], 'E': [], 'F': []
}
dfs(graph, 'A', set())
```

Sample Output

```
A B D E C F
```

Result: The program demonstrates the stated Artificial Intelligence problem.